

Software Engineering

CE2006/CZ2006

UML Review 2

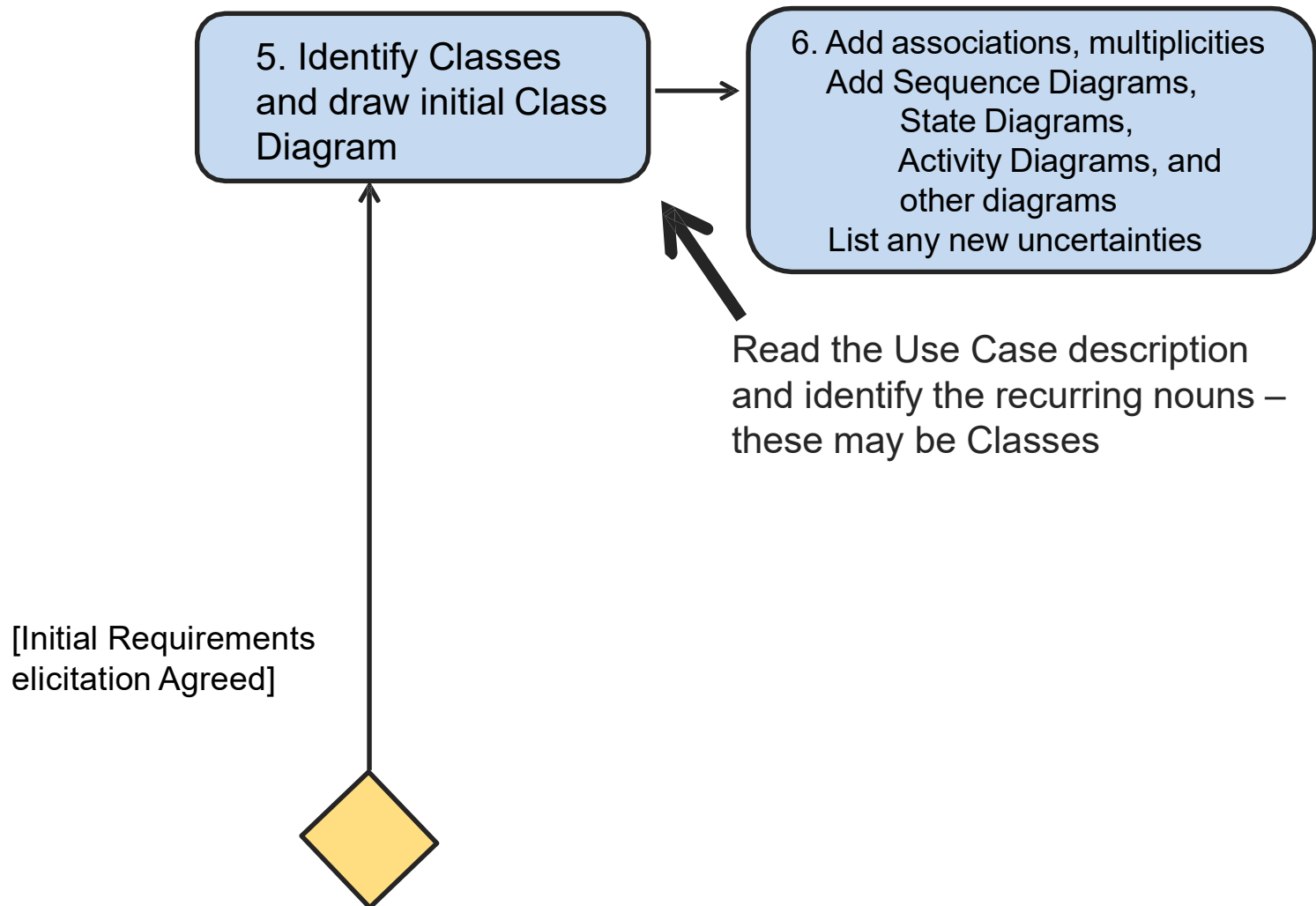
Lesson Objectives

Review UML with an example

At the end of the lesson, you should be able to:

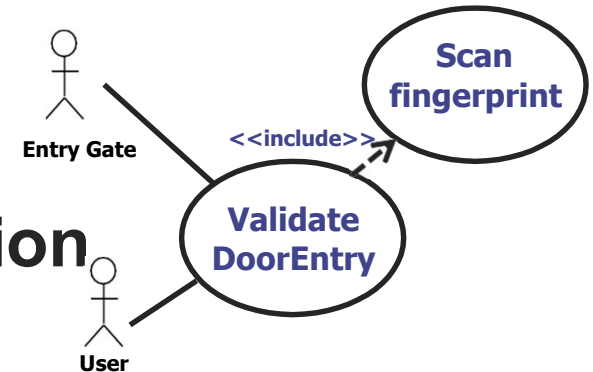
- Describe UML diagrams learned so far
- Explain the purposes of different kinds of UML diagrams
- Use different diagrams at different steps in software requirement elicitation and analysis process

Second Step – Generate Class Diagram



5. Identify Classes

ValidateDoorEntry Use Case Description



1. Employee presses **finger** on **FingerprintReader**.
2. **FingerprintReader** scans fingerprint (using use case Scan fingerprint) and sends **image** to the **FacilityAccessManager**.
3. The **FacilityAccessManager** validates the **fingerprint**.
4. If **fingerprint** is valid display the message "Please enter" and unlock the **EntryGate**.
 - 4.1 The **EntryGate** sends a signal to the **FacilityAccessManager** that it has opened and closed.
 - 4.2 The **FacilityAccessManager** locks the **EntryGate**, and logs the entry time and name of the employee who entered in the **entry log**.

AF-S4:

If **fingerprint** is not valid display the message "Access denied" and log the attempted entry time in the **entry log**.

5. Draw initial Class Diagram

INITIAL CLASSES AND TYPES



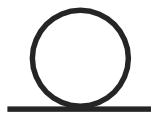
Fingerprint
Reader



EntryGate



FAMgr



Fingerprint



Entry Log

6. Add properties and operations

ADDINIG INITIAL OPERATIONS TO CLASSES



Fingerprint
Reader Interface



EntryGate Interface

Unlock()
Lock()



FASMgr

Validate(Img)
GateOpened()



Fingerprint

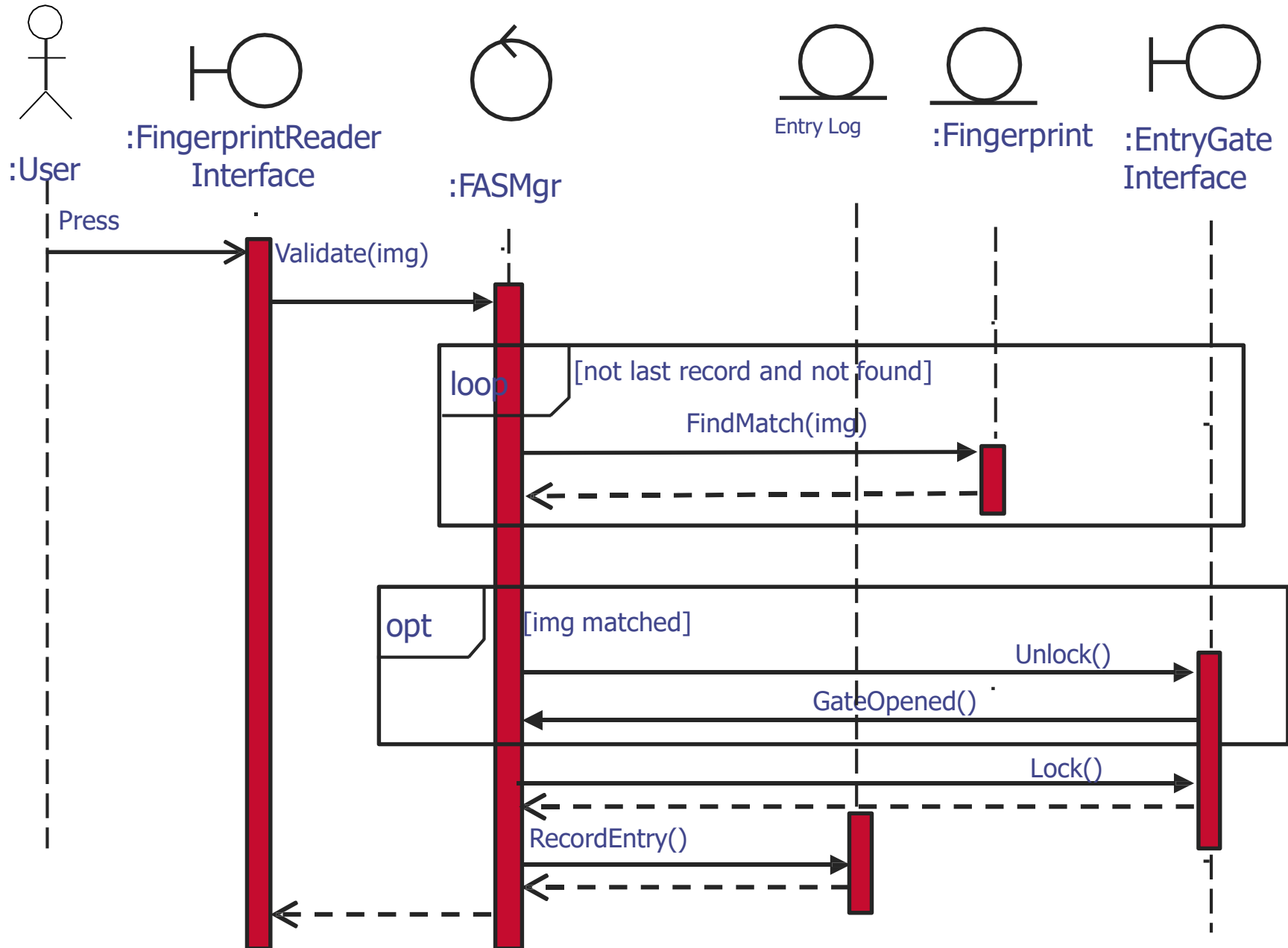
FindMatch(Img)



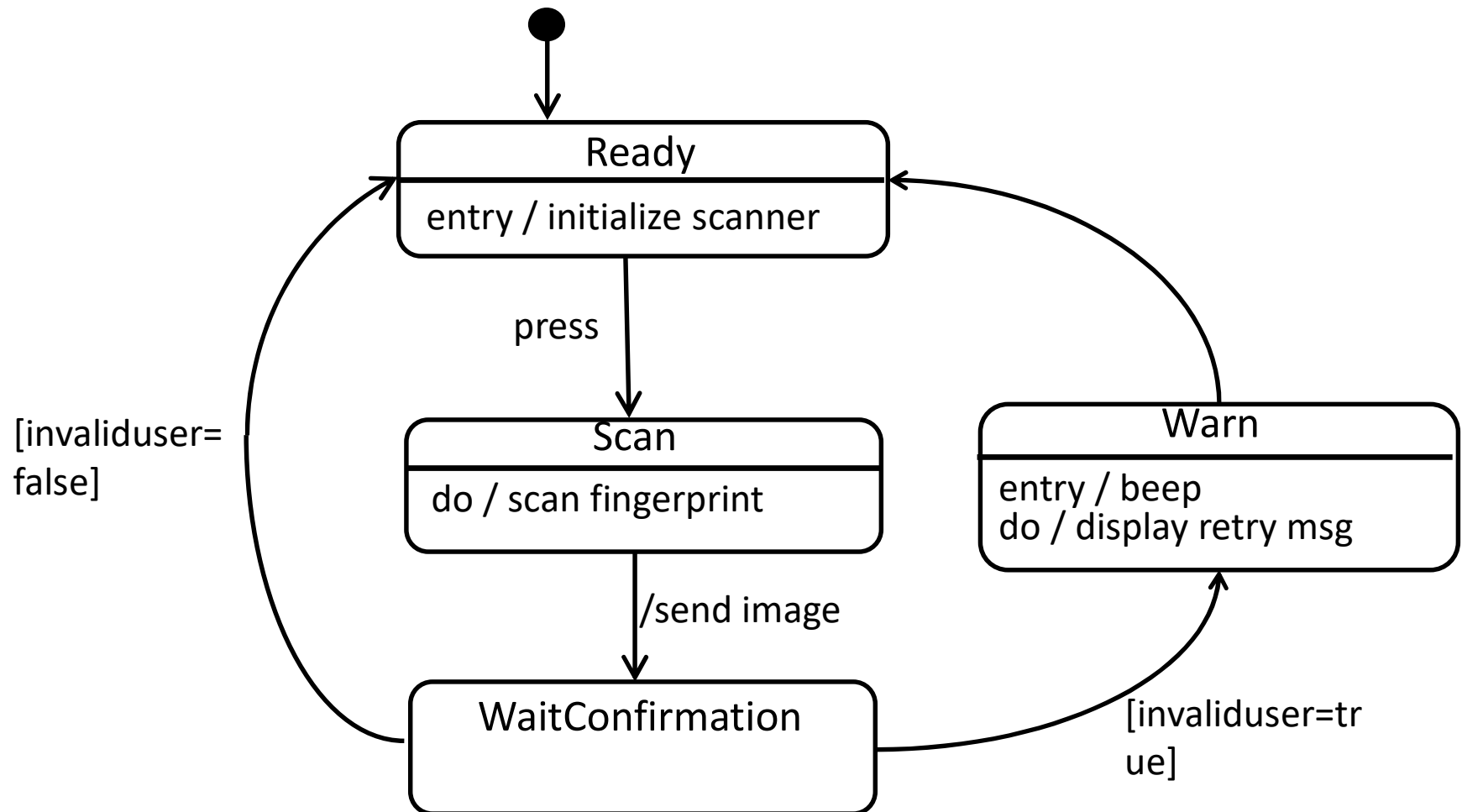
Entry Log

RecordEntry(time: Date)

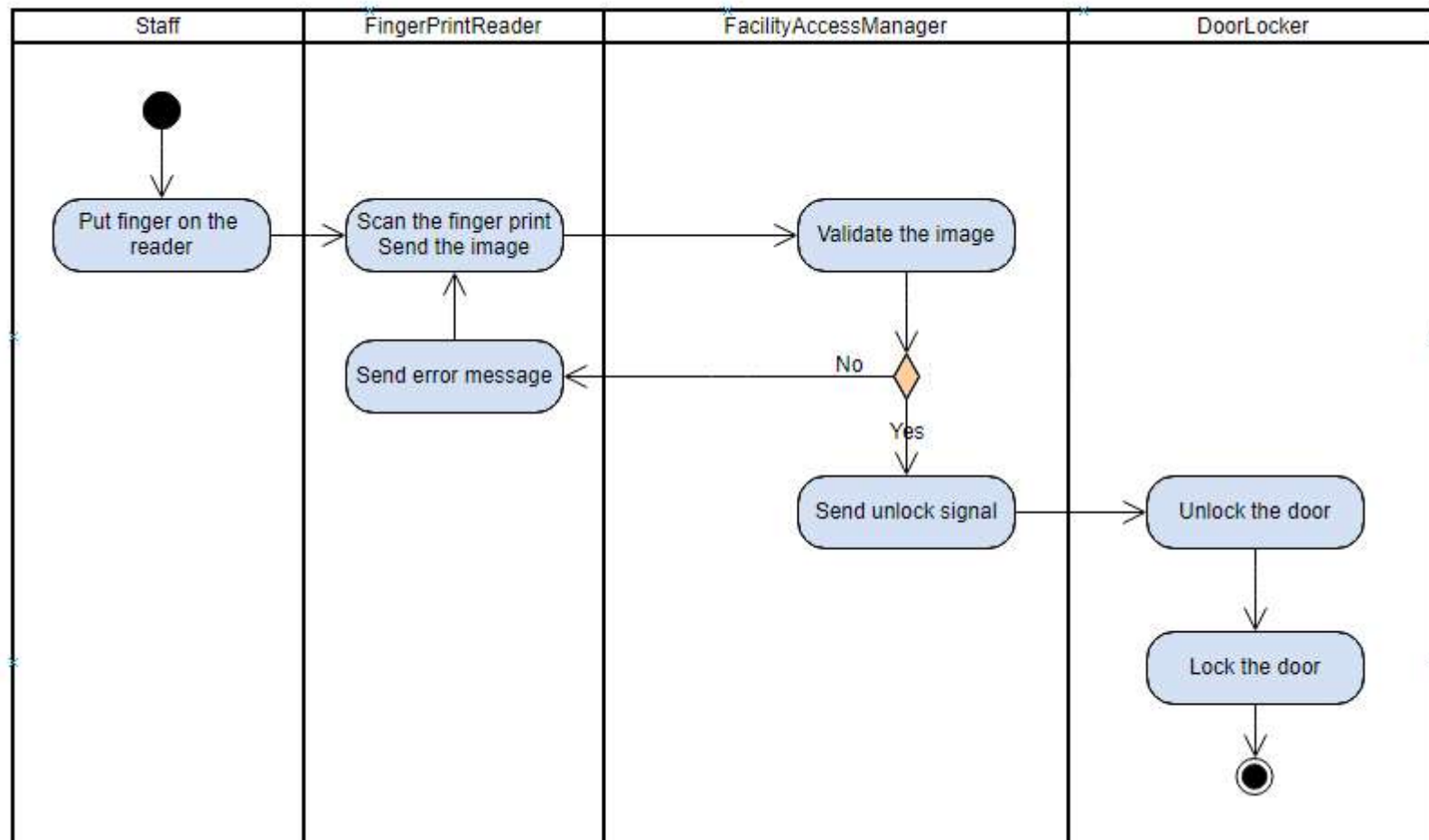
6. Add Sequence Diagrams



Example: FingerprintReader Functionality



Example: Validate Door Entry System

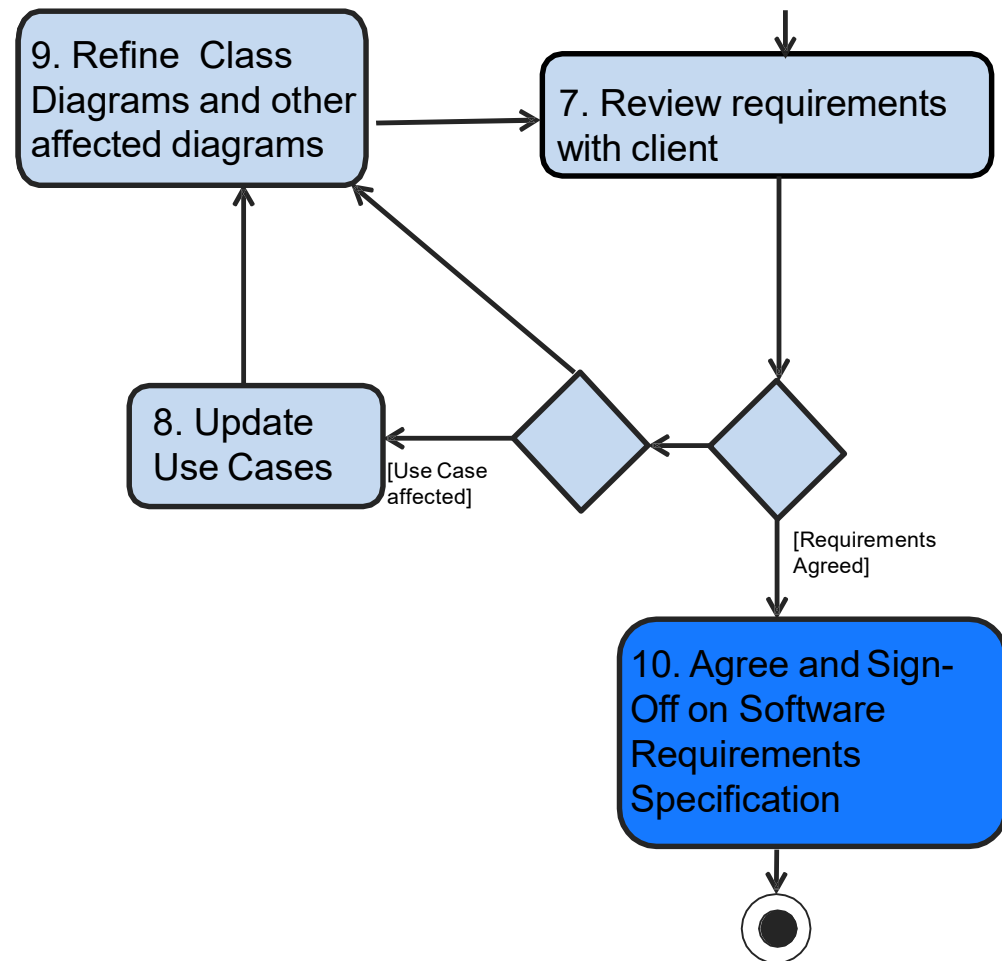


Third Step – Refine Class (& Use Case) Diagrams

Iteratively refine the Class Diagram and Sequence Diagram.

This may involve some refinement of the Use Cases.

Repeat until the Functional Requirements of the System are fully specified.



Summary

UML Review with an Example

First Half of the Course

- Introduction to Software Engineering

- Requirement Elicitation

 - Use Case Diagram

- Requirement Analysis

 - Class Diagram

 - Sequence/Communication Diagram

 - State Machine Diagram

 - Activity Diagram

Second Half of the course

- Software Design

- Software Testing